

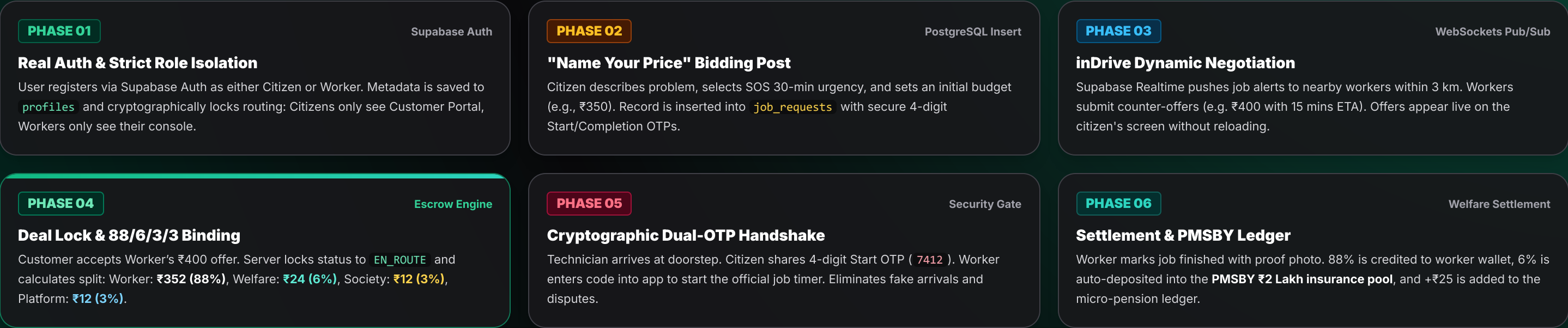
SahakarSetu (सहकार सेतु) | Technology Stack & Architectural Justifications

Cooperative-Owned Digital Service Marketplace with inDrive-Style Dynamic Price Negotiation

TECHNOLOGY MODULE	TECHNOLOGY SELECTED	WHY WE CHOSE THIS SPECIFIC TECH (ARCHITECTURAL RATIONALE)
● Client Frontend Tier	React 18 + TypeScript + Vite	Type-Safety & Sub-Second Loading: Eliminates runtime crashes across multi-step booking flows; Vite ensures ultra-fast PWA bundle sizes ($<1.2\text{s}$ FCP) on 3G/4G field mobile networks.
● Persistence & Spatial DB	Supabase (PostgreSQL + PostGIS)	ACID Escrow Reliability: PostgreSQL enforces strict transactions preventing race conditions in the 88/6/3/3 split; PostGIS indexes allow sub-second 3 km radius queries (<code>ST_DWithin</code>).
● inDrive Realtime Engine	Supabase Realtime (WebSockets)	Instant Counter-Bidding: Pushes live job offers and worker price bids in $<100\text{ms}$ across multiple concurrent devices without heavy server polling.
● Security, Auth & Isolation	Supabase Auth (JWT + RLS)	Strict Role-Based Access Control: Cryptographically locks customers into their portal and workers into their console. Row-Level Security (RLS) protects Aadhaar & phone PII under the DPDP Act.
● Accessibility Interface	Native Web Speech AI API	Zero-Cost Shramik Voice AI: Runs natively in client browsers with zero API costs; speaks job briefs aloud in Hindi & English and takes voice acceptances (<code>"स्वीकार करें"</code>).

End-to-End Technical Workflow: From Login to Settlement

Step-by-step transaction lifecycle through Supabase PostgreSQL, Realtime Bidding, and Dual-OTP verification



System Architecture & Evaluator Defense Strategy

Architectural diagram and concise, evidence-based answers to key evaluation questions

3-TIER CLOUD ARCHITECTURE

1. PRESENTATION LAYER (React 18 PWA)

Citizen Portal • Shramik Worker Console • Admin Board

Web Speech AI Voice Engine (Hindi/English)

▼ (WebSockets & REST APIs) ▼

2. CLOUD PERSISTENCE & REALTIME (Supabase Cloud)

PostgreSQL ACID Database (profiles, job_requests, bids)

Realtime Channel Pub/Sub for sub-100ms multi-device bidding

Row-Level Security (RLS) & JWT Token Verification

▼ (National Integration Hooks) ▼

DigiLocker KYC

Skill India NSDC

BharatQR / UPI

Q1: Why use Supabase instead of a custom Node server?

Answer: Supabase provides production-grade **PostgreSQL ACID compliance**, built-in **Row-Level Security (RLS)**, and **Realtime WebSockets** out of the box. This prevents financial race conditions during escrow calculations and allows our inDrive bidding engine to scale across thousands of concurrent devices in $<100\text{ms}$.

Q2: Why inDrive negotiation over fixed Urban Company rates?

Answer: Fixed pricing leads to on-site surprise quotes and job refusals. Dynamic bidding creates a **fair market price**: citizens set their budget, and nearby verified workers compete on price and arrival speed. Once agreed, the price binds an immutable 88/6/3/3 split with zero hidden platform cuts.

Q3: How do you guarantee safety inside citizen homes?

Answer: Two-tier trust: DigiLocker Aadhaar KYC + Skill India trade certification endorsed by a legal, brick-and-mortar Cooperative Society. Cryptographic Start OTP prevents fake technician check-ins, and in-app masked calling protects customer privacy.